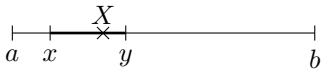
Probability 1 (MATH 11300) sample lecture – short notes

February 2021

The Uniform distribution

Given are two reals, $a < b$, and we want to define a real random variable (i.e., a random number) X that is as uniformly random as possible. Where do we start? We could say something like "X is equally likely to take each value in (a, b)". The problem with this is that such a probability is zero for any particular number: $\mathbb{P}\{X = x\} = 0$ whenever $a \le x \le b$; not very informative.

It is more useful to talk about probabilities of intervals. Take $a \le x < y \le b$, and ask about $\mathbb{P}\{X \in (x, y)\}$:



Uniformly distributed means that this probability should not depend on *where* the interval (x, y) is, only on *how long* it is. We can take this as the definition of the Uniform(a, b) distribution. It is then clear that the probability must be the ratio of the lengths of (x, y) and (a, b):

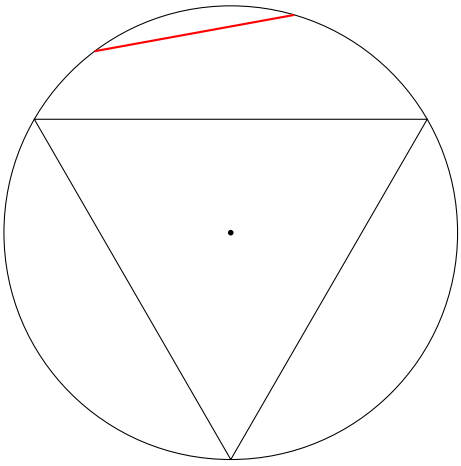
$$\mathbb{P}\{X \in (x, y)\} = \frac{y - x}{b - a}.$$

As a special case, we could ask about something called the *cumulative distribution function*:

$$F(y) := \mathbb{P}\{X \le y\} = \mathbb{P}\{X \in (a, y)\} = \frac{y - a}{b - a} \qquad \text{for } a \le y \le b,$$

while $F(y) = \mathbb{P}\{X \le y\}$ is zero when $y < a$ and one if $y > b$.

Example: Bertrand's paradox. A random chord of a circle is chosen. What is the probability that it is longer than a side of the inscribed equilateral triangle?



Expected value. To calculate the *expected value* of X, that is the average position it will take, we could break up the interval a, b into n segments (x_i, x_{i+1}), $i = 0 \dots n-1$ of equal length, and weight the position of a point, say x_i, of the i^{th} segment with the probability that X falls in that segment:

$$\mathbb{E}\, X \simeq \sum_{i=0}^{n-1} x_i \cdot \mathbb{P}\{X \in (x_i, x_{i+1})\} = \sum_{i=0}^{n-1} \Big(a + \frac{i(b-a)}{n}\Big) \cdot \frac{\frac{b-a}{n}}{b-a} = \frac{a + a + \frac{(n-1)(b-a)}{n}}{2} \cdot n \cdot \frac{1}{n} \to \frac{a+b}{2}$$

as $n \to \infty$. Alternatively, with $x \simeq \frac{i(b-a)}{n}$ and then $z = x + a$, the last sum is the approximating sequence for

$$\int_0^{b-a} (a + x) \frac{\mathrm{d}x}{b-a} = \int_a^b z \frac{\mathrm{d}z}{b-a} = \frac{z^2}{2(b-a)}\Big|_{z=a}^{z=b} = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}.$$

This is the midpoint of the interval (a, b), which is of course not surprising just by symmetry.

The factor $\frac{1}{b-a}$ in the integral is called the *(probability) density (function)* of this distribution. In a similar way, the expected value of a function g of X is calculated by $\mathbb{E}\, g(X) = \int_a^b g(z) \frac{\mathrm{d}z}{b-a}$, provided that the integral exists.